SHANMUKHA VARMA PENMETSA

Physical Design | Logical Synthesis | Static Timing Analysis | Cadence Genus, Innovus

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Methodical, curious ECE graduate who turns timing reports into design decisions. Hands-on with the Cadence Genus and Innovus flow, from RTL synthesis through a basic physical implementation, backed by SDC constraints and static timing analysis, with a habit of iterating until a design meets its goals. Automates repetitive EDA work with Tcl and Python. A fast, self-directed learner whose aim is to grow into a Physical Design engineer who delivers timing-clean, area- and power-efficient silicon.

TECHNICAL SKILLS

EDA Tools: Cadence Genus, Cadence Innovus
ASIC Design Flow: RTL Synthesis, Technology Mapping, Floorplanning
Core Competencies: Static Timing Analysis (STA), SDC Constraints, Setup/Hold Analysis, Timing Closure, QoR Analysis, Clock Gating, Timing Reports
HDL / Scripting / OS: Verilog HDL, Tcl, Python, Shell/Bash, Linux
Fundamentals: Digital Design, CMOS Fundamentals

SELECTED PROJECTS

AES Core: RTL Synthesis and Timing Analysis *Cadence Genus | 1.0 ns (1 GHz) single clock*

- Synthesized an AES core from RTL, closing timing with zero violating paths and 0.0 TNS.
- Positive slack in every timed group: r2r +0.1 ps, i2r +1.6 ps, clock-gating enable +66.8 ps, r2o +93.2 ps.
- QoR: 9,474 leaf instances (532 sequential, 8,942 combinational), 5,774 cell area, max fanout 401.

CVA6 Processor: Synthesis and Timing Analysis *Cadence Genus | 1.7 ns (∼588 MHz) | Synthesis handoff*

- Synthesized the CVA6 RISC-V processor: 178,799 leaf instances (23,675 sequential, 155,124 combinational).
- Accepted for synthesis handoff: worst slack −34 ps (2% of the clock period); TNS −10.2 ns over 510 paths.
- Traced violations (i2r, r2o, clock gating) to I/O constraints and 20% clock uncertainty; other groups meet timing.

Half-Cycle Timing Constraint Test Case *Cadence Genus | STA / SDC | Three clocks*

- Built a three-clock test case (1 ns, 2.2 ns, 4.4 ns; 1 GHz, 450 MHz, 230 MHz) to study half-cycle timing paths and their effect on setup/hold slack.
- Compared timing reports across multiple I/O delay budgets to quantify their impact on reported violations.

VLSI TRAINING

VLSI Physical Design Trainee, Sumedha IT June 2026 to Present

- Hands-on Cadence Genus workflows: RTL and library setup, elaboration, SDC application, technology mapping, optimization, and QoR review.
- Executed a basic ASIC flow in Cadence Innovus, extending synthesis work into physical implementation.
- Writing Tcl and Linux scripts to automate EDA tasks and report analysis; scripts published in the GitHub repository tcl-scripts.

EDUCATION

B.Tech, Electronics and Communication Engineering September 2022 to April 2026
Vignan's Institute of Information Technology **GPA: 9.01/10**

ADDITIONAL EXPERIENCE

Embedded Systems Intern, Tecky Bot May 2024 to June 2024

- Programmed microcontrollers in Embedded C using Arduino and interfaced them with external hardware.
- Built an Arduino Uno school bell automation system using time-controlled logic to trigger the bell on schedule.

CERTIFICATIONS

Cisco Networking Academy: CCNA Fundamentals; Introduction to Cybersecurity; Introduction to Operating Systems
NPTEL: Programming and Data Structures in Python | edX: Introduction to IoT and Cloud Computing